

**ExcaVision: An IoT-based and AI-assisted continuous monitoring system for
temporary excavation shoring**

An Application Development Project
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FAUSTORILLA, GERALD

JUMAWAN, MIGUEL LUIS

MANZANILLA, JOEY JR.

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I. Introduction

A. Overview

Shoring systems support the walls of deep excavations during foundation work. Unstable shoring is a major danger in construction. Even small movements in shoring walls can go unnoticed by site engineers, which might lead to excavation collapse—a sudden failure of the excavation wall or support system that can cause injuries or project delays. The usual approach relies on well-planned shoring systems, regular inspections, and the skill of the site engineer.

Although these procedures usually work, interviews with contractors and engineers in Siquijor and Bacolod showed that excavation conditions can change a lot between scheduled inspections, especially during heavy rain.

Their current process relies on scheduled manual inspections, and both groups said that having continuous monitoring between inspections would be helpful. ExcaVision is an experimental IoT-system allowing live monitoring for shoring systems. It uses tilt sensors attached to the shoring to continuously monitor movement. The sensor data is sent to a controller and displayed in a mobile app, so the Site Engineer can see the status, layout and trends. The AI component utilizes tilt readings over time to identify hazardous movement and issues an early advisory prior to the tilt limit established by the engineer being reached.

B. Research Problem

The project tackles the common problem of subtle shoring movement at deep excavation sites and the lack of tools available for site engineers to monitor shoring between inspections. It is important for site engineers to quickly detect any abnormal movements especially when there is change in the weather conditions. Ongoing monitoring of shoring walls is needed as they can weaken slowly between inspections.

C. Research Objectives

The aim is to develop a prototype for real time monitoring of excavation shoring system using IoT sensors. ExcaVision is a preventive system that uses AI to give early warnings when movement increases. The specific objectives are:

- Design and build a proof of concept that uses multiple tilt sensors for continuous monitoring of shoring systems.
- Build a mobile app where the site engineer will view sensor status, site overview, alerts, and tilt history through a simple, clean interface.
- Add a simple AI feature that uses historical sensor data and gives an early advisory before a tilt limit is exceeded.
- Show, through a functioning prototype, that continuous monitoring can boost but not substitute for scheduled maintenance inspections.
- Build climate resilience and adaptive capacity in support of SDG 13.1 as heavy rain increases the risk of excavation hazards;

D. Scope and Limitations

The study will be conducted from July to October 2026. The software is based on a miniature excavation model, which closely simulates a real shoring system. The application will be used to focus on real time tilt monitoring, issuing warning alerts and early advisories based on the past sensor data to help in decision making and create awareness. System shall be tested by Site Engineers , Contractors or other qualified professionals .

This study will not be verified in a real construction site, so it may not fully correspond to the actual structure and soil conditions of an actual excavation. The accuracy of the prediction depends on the quality of the sensor data of the prototype. It is important to be clear that this system does not replace regular site inspections, cannot predict structural failure, and does not guarantee worker safety by itself. However, this proof-of-concept shows that using IoT monitoring with AI alerts is a practical approach for excavation projects.

E. Review of Related Literature

IoT-Based Monitoring for Construction Shoring:

Su et al (2018) investigates the feasibility of an IOT-based monitoring system to determine if the scaffolding is faulty (moving, bending, tilting, or carrying too much weight). The monitoring system uses wireless displacement sensors, pressure sensors, and tilt-angle sensors for the monitoring of scaffolding.

Their monitoring of scaffolding movement has showed great possibility for using wireless sensor network to continuously monitor shoring movement, this inspires the use of sensor nodes and central-control-unit architecture in ExcaVision that monitoring the shoring behavior for excavation retaining structure, although shoring in this context has a totally different purpose from the scaffolding one.

Tilt Monitoring for Excavation-Adjacent Structures:

Industry guidance suggests the use of tilt sensors at the top of retaining or shoring walls as a response to the indication that any movement is usually first observed at this location and for that reason many industry professionals use wireless networks of IoT tilt sensors to report tilt to a data repository at a specified frequency with higher frequency data collection automatically triggered once the movement rate approaches certain values, as is represented by ExcaVision's variable frequency monitoring concept (Oculus Monitoring, 2025). ExcaVision uses this as a reference for the placement of sensors in the pilings for data collection.

AI-Assisted Early Warning and Trend Detection:

Other than sending alerts when tilt readings exceed safety limits, literature in Structural Health Monitoring has started using prediction tools. These tools look at collected data to spot problems before they happen. The authors of Encardio-Rite (2025) report that the newest monitoring systems do more than trigger alerts at set safety limits. They can also forecast by watching how sensor readings behave differently from their normal patterns. ExcaVision uses the same idea as it sends out an early advisory alert when the tilt reading is getting close to the safety limit set by the engineer. It then issues a warning alert if the safety limit is exceeded.

II. Project Organization (Team Description)



Figure 1.1.
Team member 1

Miguel Luis Jumawan

AI & Backend Developer
/ Team Coordinator



Figure 1.2.
Team member 2

Gerald Faustorilla

App, AI & Backend
Developer



Figure 1.3.
Team member 3

Joey Manzanilla Jr.

Hardware Developer

Miguel Luis Jumawan: *AI & Backend Developer / Team Coordinator*

Focuses on AI and backend development, supports other areas of the project when needed, and coordinates the team.

Gerald Faustorilla: *App, AI & Backend Developer*

Develops the mobile app, works on AI trend analysis features, and manages backend tasks. He is also studying Arduino and hardware basics to assist with hardware development when needed.

Joey Manzanilla Jr.: *Hardware & Backend Developer*

Manages Arduino and hardware tasks for the sensor nodes and head sensor node. He also helps integrate sensor node readings into the backend.

III. Software Process Model Description

ExcaVision's development covers hardware, embedded firmware, a mobile app, and AI trend analysis. The Incremental Model works the best, as the team is still learning in some of these areas. That lets the team build and test in each stages. It begins with a single sensor node and a simple app interface. Then, add more sensors, views and functionalities. And finally add alerting and AI advising capabilities. This approach helps to catch problems early and prevent severe setbacks later on. And there is feedback and validation at each level which is more suited to the experience of the team and fixed time frame of the project than a rigid Waterfall."

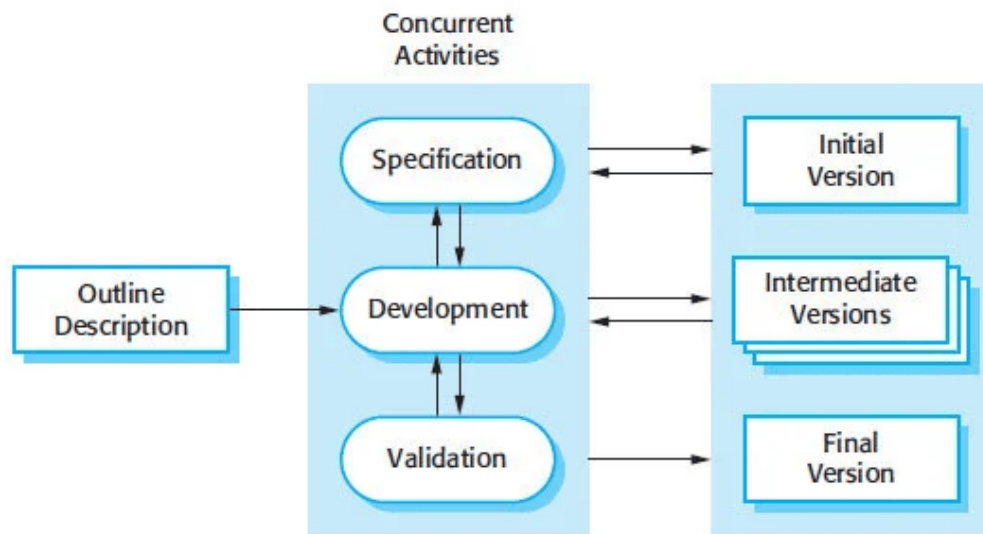


Figure 2. Incremental Model Diagram

1. Initial Version

In this phase the team builds and tests a basic mobile dashboard and one sensor node.

Activities/Tasks:

- Build 1 tilt sensor node
- Code to read the tilt data and send it
- Create a simple app screen for raw sensor data

Deliverables:

- One working sensor prototype
- ExcaVision app first version

2. Intermediate Versions

Now the team connects multiple sensor nodes to its corresponding head sensor node and adds site overview screens to the app.

Activities/Tasks:

- Build the head sensor node and connect a number of sensor nodes via RS-485.
- Develop the app site details, layout and live sensor screens

Deliverables:

- Several sensor nodes connected and working with the head sensor node.
- Updated application, new screens and views.

3. Final Version

In this phase the team will finalize the alerting system, develop the AI trend-analysis feature and complete the project.

Activities/Tasks:

- Create alert list, establish history or trend tracking, adjust threshold settings, create push notification screens.
- Apply AI trend-analysis logic and issue early advisories.
- Integrate all modules and run full end-to-end system tests.
- Fill in the documents like SDP, SRS, SDD and test results and present the project.

Deliverables:

- Final ExcaVision proof-of-concept
- Finalized Documents

IV. Risk Analysis

This section describes the risks to ExcaVision and ways to ensure that the project is delivered on time, within scope and of presentable quality.

Table 1. Risk Analysis

Risk Description	Category	Probability	Impact	Response Strategy	Management / Mitigation Strategy
Sensor readings drift or get noisy, causing false alerts.	Technical Risk	Medium	Medium	Mitigate	Calibrate tilt sensors before demonstration; apply basic smoothing/averaging to raw readings; test thresholds against controlled movements on the miniature model.
RS-485 communication failure between sensor nodes and its corresponding head sensor node.	Technical Risk	Medium	High	Mitigate	Test each daisy-chained connection individually before full integration; keep cable runs short; add a visible “sensor offline” status in the app
Members may be unfamiliar with the tools used in hardware, software, and	Resources and Team Risk	High	Medium	Mitigate	Assign a person to lead each increment, even if roles are not hard and fast. Check in often so

AI development.					if someone is stuck they can get help quickly. Put a time limit on unfamiliar tasks and pass to whoever knows it best if stuck.
The AI might mistake normal sensor noise for an actual increasing trend.	Technical Risk	Medium	Medium	Mitigate	Validate the trend-analysis logic against recorded sensor data from controlled test movements; keep the AI as an advisory layer only.
The miniature demonstration model behaves differently from full-scale excavation shoring.	Technical Risk	High	Low	Accept	Clearly document the prototype as a proof-of-concept, not a field-validated system.
Site Engineers lose trust in the system if alerts are frequent, unclear, or unexplained.	User Acceptance Risk	Moderate	High	Mitigate	Show supporting history/trend data alongside each alert; clearly distinguish Early Advisory from Warning Alert; make thresholds configurable by the engineer.

V. Hardware and Software Resource Requirements

1. Hardware

Table 2.1. Hardware requirements (End-user)

Components	Specification
Device Type	Android smartphone or tablet
CPU	Quad-core 1.5Ghz or better
RAM	2 GB
Storage	500 MB free
Screen Size	5.0” or larger
Connectivity	Wi-Fi or mobile data (for push notifications and app sync)

Table 2.2. Hardware requirements (Developers)

Components	Specification
CPU	Intel Core i5 / AMD Ryzen 5 or better
RAM	8 GB
Storage	256 GB SSD
Device Type	Laptop / Desktop
Tilt Sensor	MPU6050
Sensor Node Microcontroller	Arduino Nano ATmega328P
Head Sensor Node Microcontroller	ESP32 dev module
I2C Multiplexer	TCA9548A
Communication Interface	RS-485 (Half Duplex)

Communication Module	MAX485 RS-485 TTL Module
Power Input	5V DC
Demonstration Model	Miniature excavation shoring model

2. Software

Table 2.3. Software requirements (End-user)

Components	Specification
Operating System	Android 8 (Oreo) or better
Permissions	Notifications, Storage
App Installation	Downloadable APK

Table 2.4. Software requirements (Developers)

Components	Specification
Operating System	Windows 10/11 or macOS
Firmware Development Environment	Arduino IDE
Mobile App Development Environment	VS Code and Android Studio
Version Control	Git + Github

3. Network

Table 2.5. Network requirements (End-user)

Components	Specification
Internet	Wi-Fi or 3G/mobile data or better, for receiving alerts and syncing sensor data

Table 2.6. Network requirements (Developers)

Components	Specification
Internet	Stable connection (10 Mbps or better) for version control, testing, and collaboration

VI. Work Breakdown (WBS, Work Breakdown Structure)

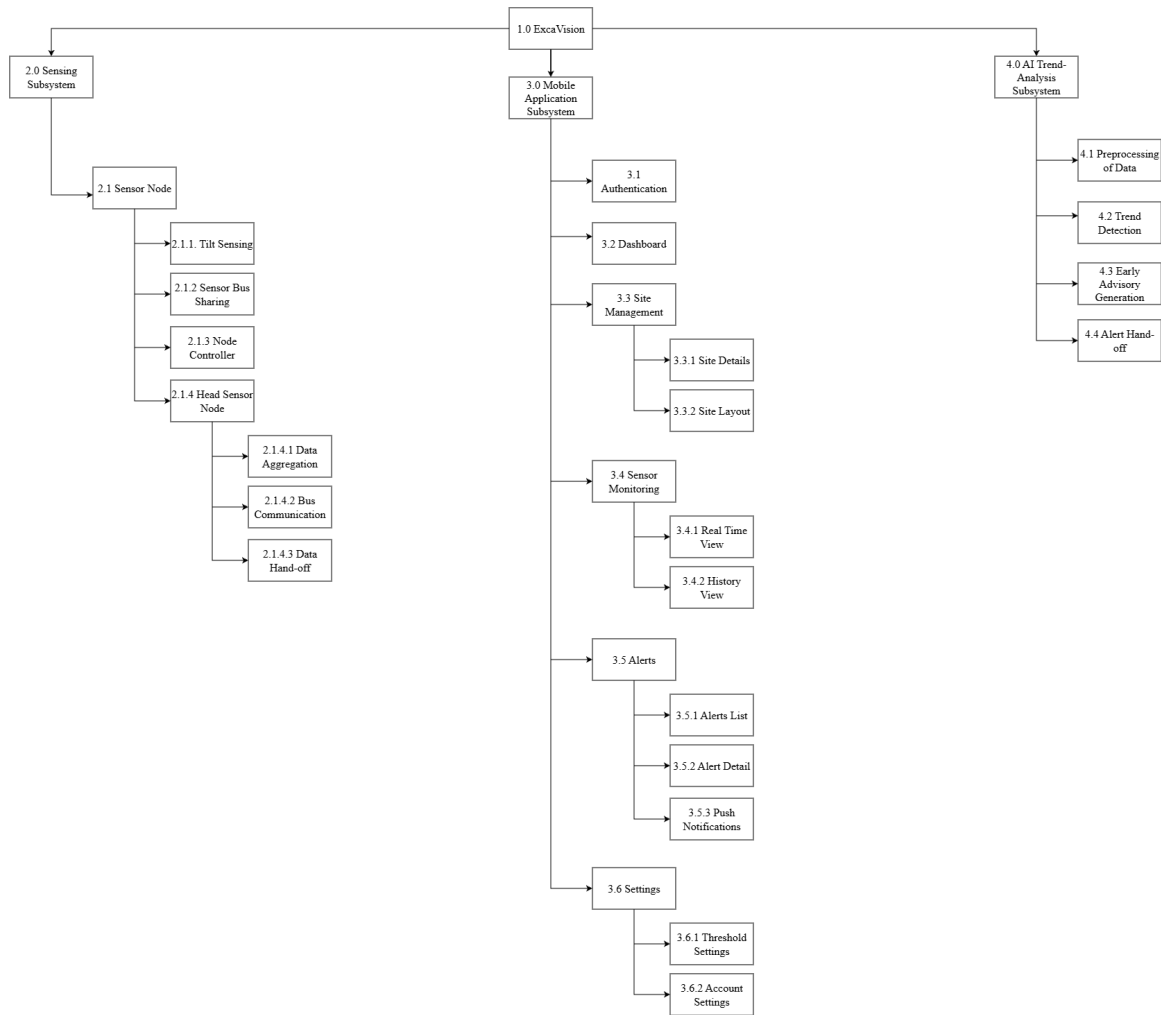


Figure 3. WBS Figure

1.0 ExcaVision

- 1.0 ExcaVision - The overall monitoring system.

2.0 Sensing Subsystem

- 2.0 Sensing Subsystem — Physical sensor network that measures shoring movement.

- 2.1 Sensor Node — Individual tilt-sensing unit installed along the shoring.
 - 2.1.1 Tilt Sensing — Reads raw tilt data from the MPU6050 sensor.
 - 2.1.2 Sensor Bus Sharing — Uses a TCA9548A multiplexer so multiple MPU6050 sensors can share a single I2C line.
 - 2.1.3 Node Controller — Microcontroller to collect data from the sensor bus, process and transmit to its corresponding head sensor node through RS-485.
- 2.2 Head Sensor Node — Collects data from all connected corresponding sensor nodes.
 - 2.2.1 Data Aggregation — The readings of the previous corresponding sensor nodes are collected and aggregated by the head sensor node.
 - 2.2.2 Bus communication — Control of bus node RS-485.
 - 2.2.3 Data Hand-off — Delivers aggregated data to the mobile app, typically using Wi-Fi or a wired connection to the host device.

3.0 Mobile Application Subsystem

- 3.0 Mobile Application Subsystem — ExcaVision app for the Site Engineer.
 - 3.1 Authentication — User login, Session management based on the user database of the app.
 - 3.2 Dashboard — A view of a site or sensor, along with recent alerts, taken from the backend database.
 - 3.3 Site Management — Information on site level and layout.

- 3.3.1 Site Details — Name, status and number of sensors, in the backend database.
- 3.3.2 Site Layout — Shows the location of the sensors on a map view inside the application.
- 3.4 Sensor Monitoring — Views of data per sensor.
 - 3.4.1 Real Time View — Shows current tilt reading from the data feed of the head sensor nodes.
 - 3.4.2 History View — Displays historical tilt readings retrieved from the backend database.
- 3.5 Alerts – Alert list, details and delivery
 - 3.5.1 Alerts List — Lists all the current and historical alerts that are queried from the backend database.
 - 3.5.2 Alert Detail — displays the complete detail of a selected alert.
 - 3.5.3 Push Notifications – Send notifications to the device of the engineer, probably with a push service like Firebase Cloud Messaging.
- 3.6 Settings — Account and threshold configuration.
 - 3.6.1 Threshold Settings — Includes early Advisory or Warning thresholds, stored per site or sensor.
 - 3.6.2 Account Settings — Allows management of engineer account details.

4.0 AI Trend-Analysis Subsystem

- 4.0 AI Trend-Analysis Subsystem — Lightweight advisory layer on top of sensor data.
 - 4.1 Preprocessing of Data — Cleans and segments historical tilt data retrieved from the backend database.
 - 4.2 Trend Detection – A lightweight algorithm for identifying consistently increasing movement trends.
 - 4.3 Early Advisory Generation — Produces the Early Advisory before the fixed threshold is reached.
 - 4.4 Alert Hand-off — Passes the Early Advisory to the Alerts section (3.5).

VII. Project Schedule

Table 3. Project Schedule

ID	Activity	Predecessor	Duration (weeks)
Preliminary Investigation			
A	Conduct target audience interviews (contractors or engineers in Siquijor, Bacolod)	—	1
B	Project concept proposal (select final idea, storyboard, draft and submit)	A	1
Development Plan			
C	Define project organization and the software process model	B	1
D	Conduct the risk analysis	B	1
E	Create the work breakdown structure and project schedule	C, D	1
Requirement Analysis			
F	Define the user and system requirements	C, D, E	1
G	Define system architecture and research	F	1

	sensor/AI components		
Design Phase			
H	Design sensor nodes and head sensor nodes architecture, design the data schema	G	1
I	Design the UI/UX wireframes & mockup for all core app modules (Dashboard 3.2, Site Mgmt. 3.3, Sensor Monitoring 3.4, Alerts 3.5, Settings 3.6, Auth 3.1)	H	1
Coding and Unit Tests			
J	Code hardware and firmware: • Sensor node • RS-485 communication • Head sensor node	I	3
K	Create mobile app core modules and implement AI trend-analysis feature	I	2
Integration & Testing			
L	Integrate hardware, firmware, mobile app, and AI feature, then perform system tests	J, K	2
M	Fixing bugs, optimization, and presentation	L	1

	preparation		
Deployment & Documentation			
N	Finalize documentation and present project	M	1

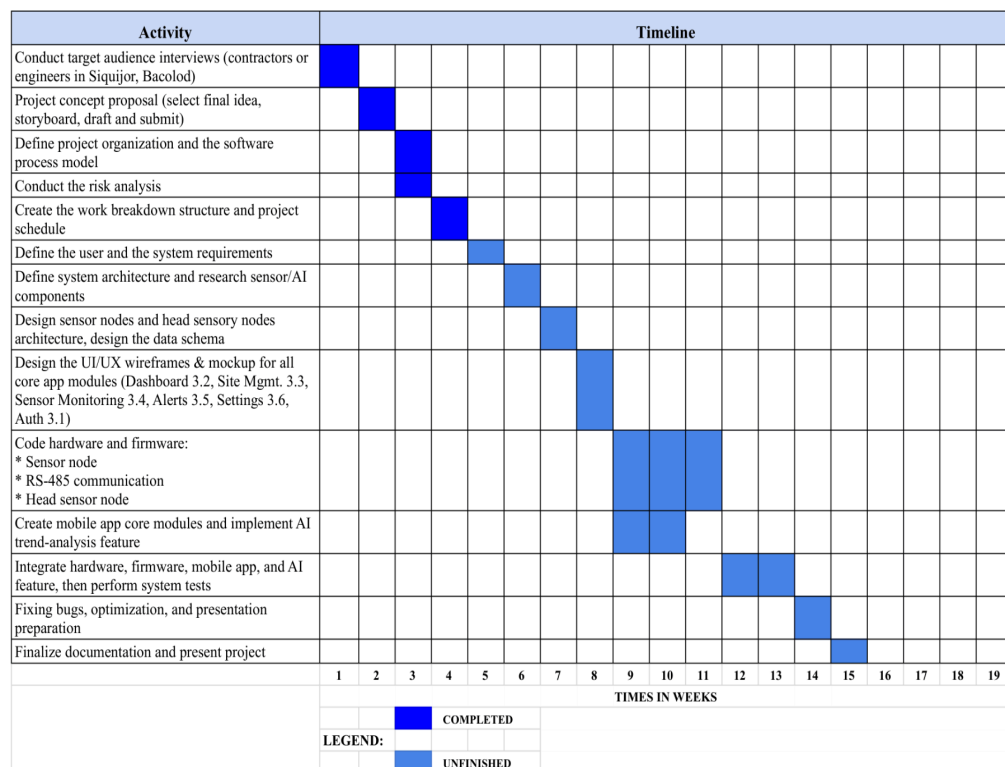


Figure 4. Gantt Chart

VIII. Ethical Considerations

1. Privacy of users

The privacy of users is of the highest importance. The interviews with the contractors and engineers will be utilized for the purpose of developing the prototype and will not be disclosed outside the development of the project. The sensor data, such as the tilt measurements, the alert logs, and the timestamps, will be used only for the general development of the prototype. The system will not store personal information about the construction workers or site visitors.

2. Licenses for software

The project will be developed in parallel with open source tools and libraries. These are the Arduino IDE and the common sensor and communication libraries. They are under Apache 2.0, MIT and GPL, which means it can be used legally. All licenses of software components are fully documented and any references, libraries or components used in the hardware or software are appropriately credited in the final documentation.

3. Data privacy and security

Sensor and account data will be protected with the use of reasonable security practices appropriate for a prototype, one example is authenticated access to the mobile applications, and no transmission of sensor data to any third parties. Access to sensor logs is only limited to the project team. Given the prototype does not currently handle payment or highly sensitive personal information, formal

compliance with regulations such as the Data Privacy Act of 2012 (Philippines) is not required at this stage, but the team will follow its principles by collecting only what is needed and being transparent with how the data will be used.

4. User impact

ExcaVision is designed to help Site Engineers by giving them earlier awareness of shoring movement. It does not replace engineers or scheduled inspections, and it does not predict structural collapse or guarantee safety. The final decision to inspect or continue excavation will always be with the responsible Site Engineer. A possible negative impact is becoming too dependent on the system or experiencing alert fatigue if advisories are frequent or unclear; to mitigate this, alerts are paired with supporting history or trend data and a clearer distinction between an informal Early Advisory and a Warning Alert, threshold will be configurable by the engineer.